

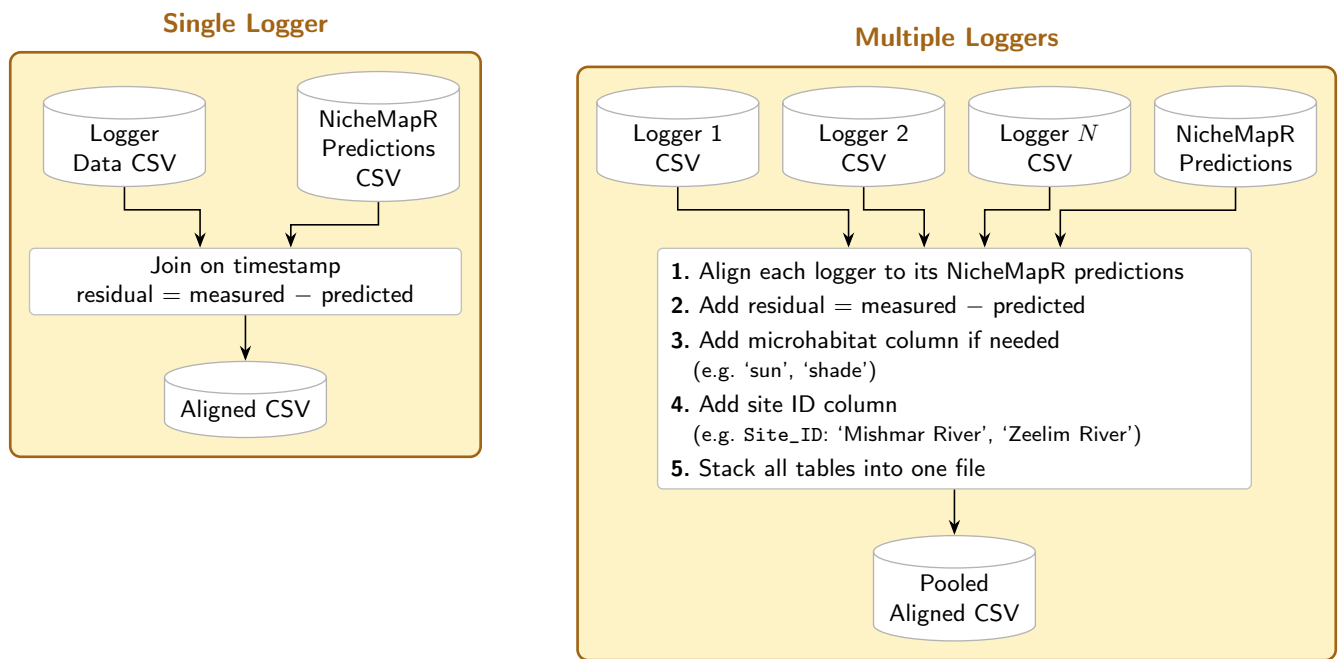


Figure 1: Data preparation users need to perform outside the `microclCorr` package. First, the user aligns the logger measurements with the NicheMapR predictions on timestamp and computes the residual (measured minus predicted). For a single logger (left), this produces a single aligned CSV file. For multiple loggers (right), the user additionally assigns a microhabitat label and a site identifier to each row, then concatenates all loggers into one file. Both paths produce an aligned CSV that serves as input to the ML pipeline (Fig. 2).

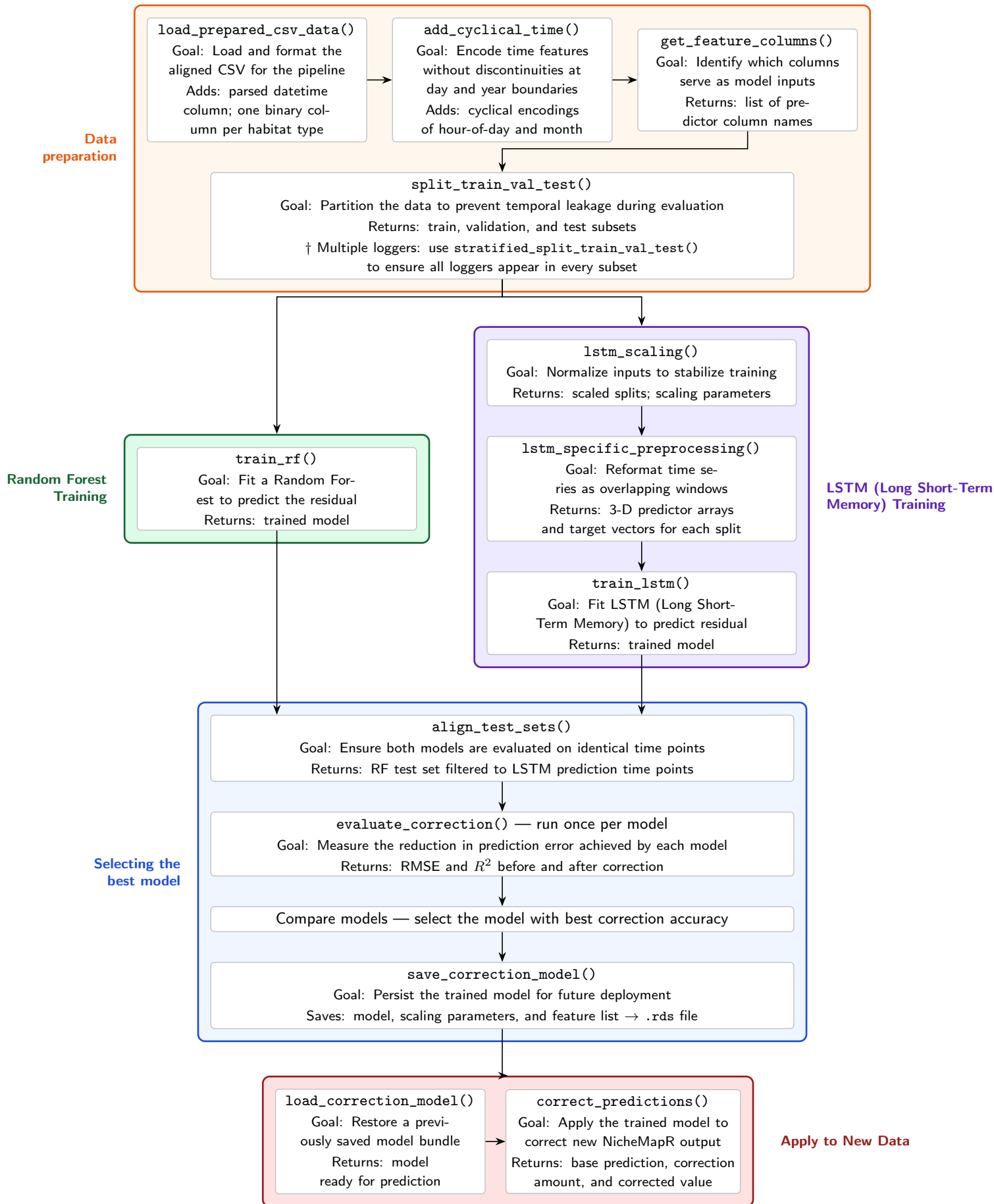


Figure 2: Machine learning pipeline implemented within `microclCorr`. First, the aligned CSV is loaded, cyclical time features are added, predictor columns are identified, and the data are split into training, validation, and test subsets. Next, the data are passed to two parallel training branches: a Random Forest (left) and a Long Short-Term Memory (LSTM) network (right); steps marked with † differ for the pooled (multiple-logger) scenario. Next, both models are evaluated on identical test time points and compared to select the best correction accuracy. Finally, the best-performing model is saved and can be applied to correct new predictions.